

✓ ASSIGNMENT 10

Name - Ayush Pathak

Roll no - 60

Batch - 03

Problem Statement : Assignment on Regression technique.

Download temperature data from the link below.

<https://www.kaggle.com/venky73/temperaturesof-india?select=temperatures.csv>

This data consists of temperatures of INDIA averaging the temperatures of all places month wise. Temperatures values are recorded in CELSIUS

1. Apply Linear Regression using a suitable library function and predict the Month-wise temperature.
2. Assess the performance of regression models using MSE, MAE and R-Square metrics
3. Visualize a simple regression model.

1. Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

2. Upload Dataset

```
from google.colab import files
uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving temperatures.csv to temperatures.csv

```
dataset = pd.read_csv('temperatures.csv')
dataset.head()
```

	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
0	1901	22.40	24.14	29.07	31.91	33.41	33.18	31.21	30.39	30.47	29.97	27.3	26.3
1	1902	24.93	26.58	29.77	31.78	33.73	32.91	30.92	30.73	29.80	29.12	26.3	26.0
2	1903	23.44	25.03	27.83	31.39	32.91	33.00	31.34	29.98	29.85	29.04	26.0	26.3
3	1904	22.50	24.73	28.21	32.02	32.64	32.07	30.36	30.09	30.04	29.20	26.3	27.5
4	1905	22.00	22.83	26.68	30.01	33.32	33.25	31.44	30.68	30.12	30.67	27.5	

3. Reshape Dataset

```
# Select only monthly columns
months = ['JAN', 'FEB', 'MAR', 'APR', 'MAY', 'JUN',
          'JUL', 'AUG', 'SEP', 'OCT', 'NOV', 'DEC']

# Convert wide → long format
df = dataset.melt(value_vars=months, var_name='Month', value_name='Temperature')

df.head()
```

	Month	Temperature
0	JAN	22.40
1	JAN	24.93
2	JAN	23.44
3	JAN	22.50
4	JAN	22.00

4. Convert Month Names to Numbers

```
month_map = {
    'JAN':1, 'FEB':2, 'MAR':3, 'APR':4,
    'MAY':5, 'JUN':6, 'JUL':7, 'AUG':8,
    'SEP':9, 'OCT':10, 'NOV':11, 'DEC':12
}

df['Month'] = df['Month'].map(month_map)
```

5. Prepare Data

```
X = df[['Month']].values
y = df['Temperature'].values
```

6. Train-Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

7. Apply Linear Regression

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
```

```
▼ LinearRegression ⓘ ?
LinearRegression()
```

8. Predictions

```
y_pred = model.predict(X_test)
```

9. Evaluation Metrics

```
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

mse = mean_squared_error(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("MSE:", mse)
print("MAE:", mae)
print("R2 Score:", r2)
```

MSE: 9.999806682283973
MAE: 2.705271869781447
R2 Score: -0.005573628993982727

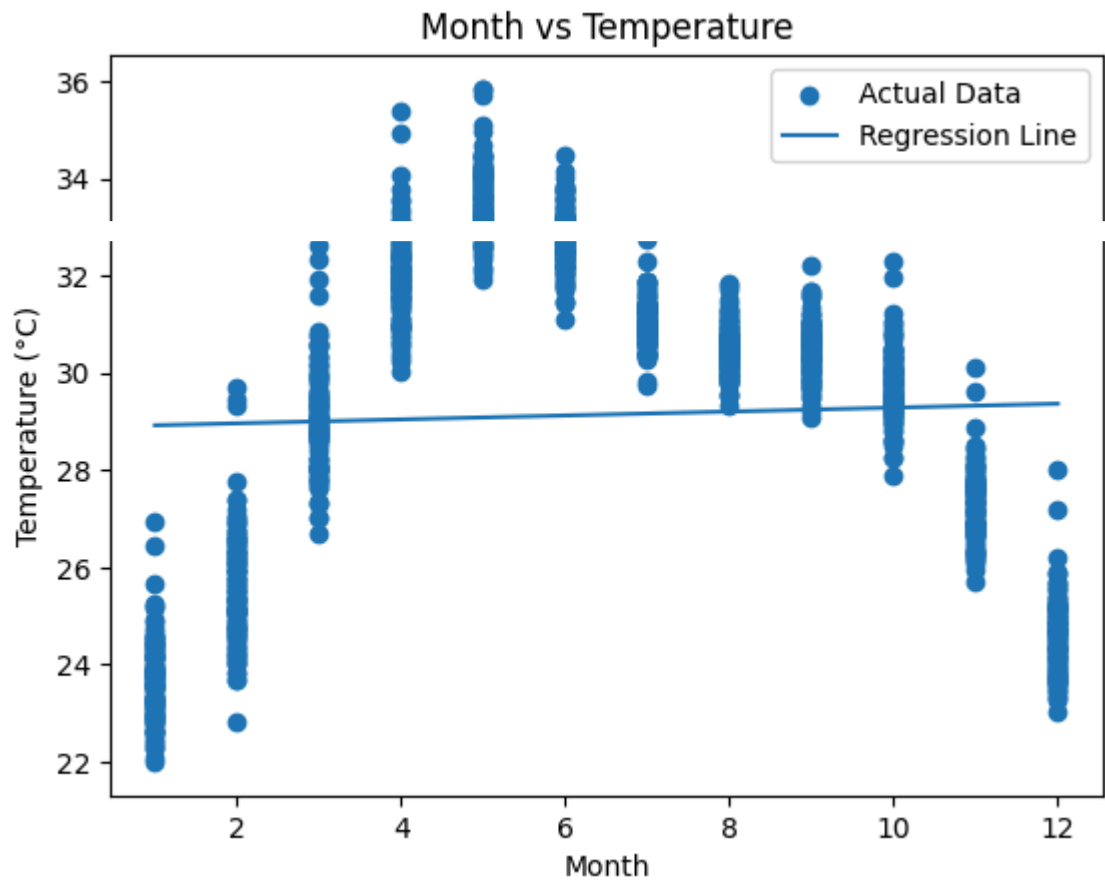
10. Visualization

```
plt.scatter(X, y, label='Actual Data')

plt.plot(X, model.predict(X), label='Regression Line')

plt.xlabel("Month")
plt.ylabel("Temperature (°C)")
```

```
plt.title("Month vs Temperature")  
plt.legend()
```



11. Predict for Any Month

```
month = np.array([[6]]) # June  
print("Predicted Temperature:", model.predict(month)[0])
```

Predicted Temperature: 29.122498288380676